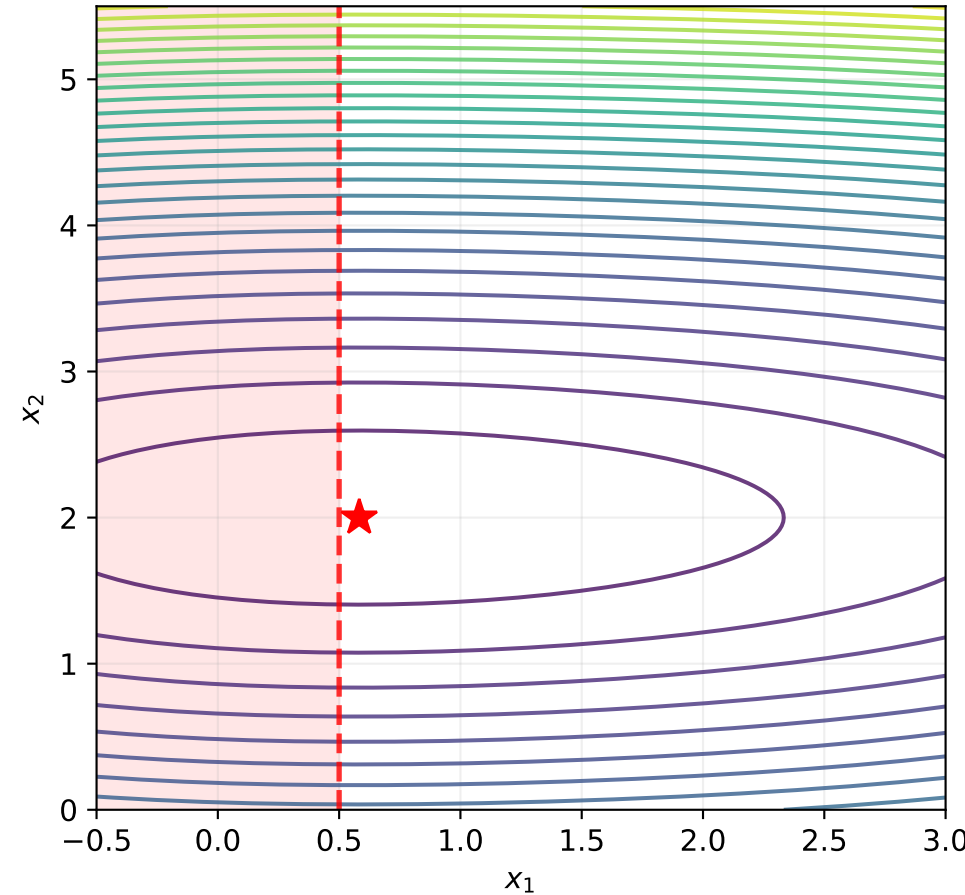
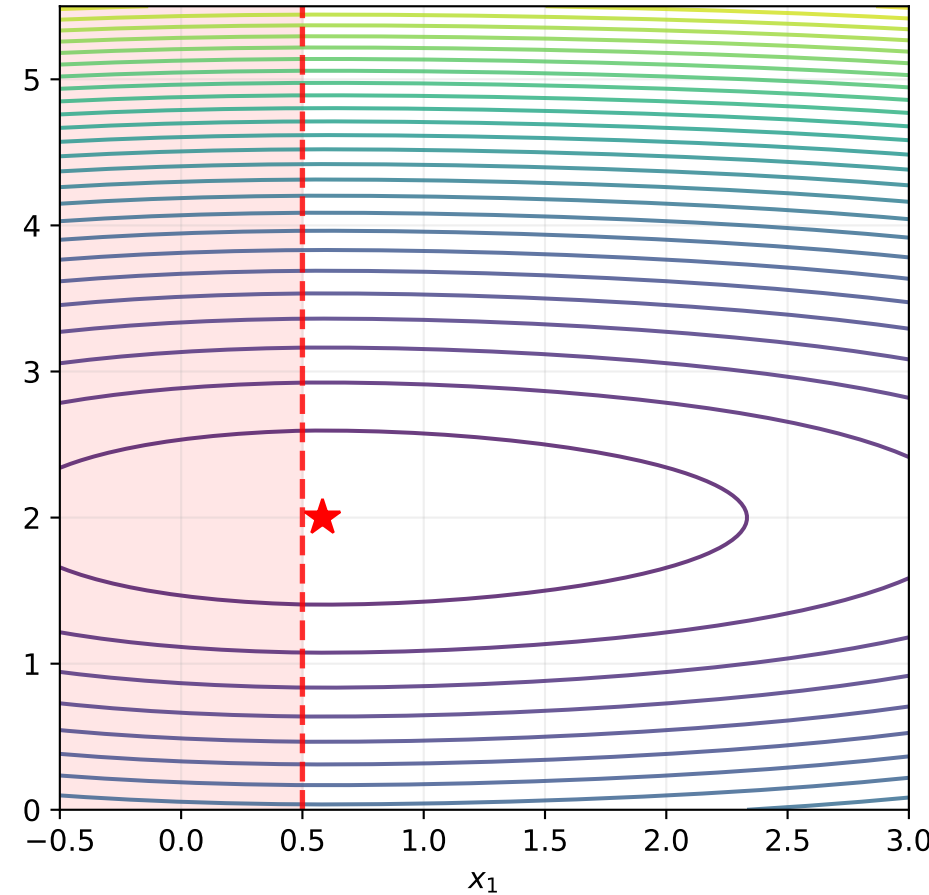


Q5: Penalised Objective $F_\lambda(x) = f(x) + \lambda \max(0, 0.5 - x_1)$ for Different λ
(red dashed = constraint boundary $x_1 = 0.5$)

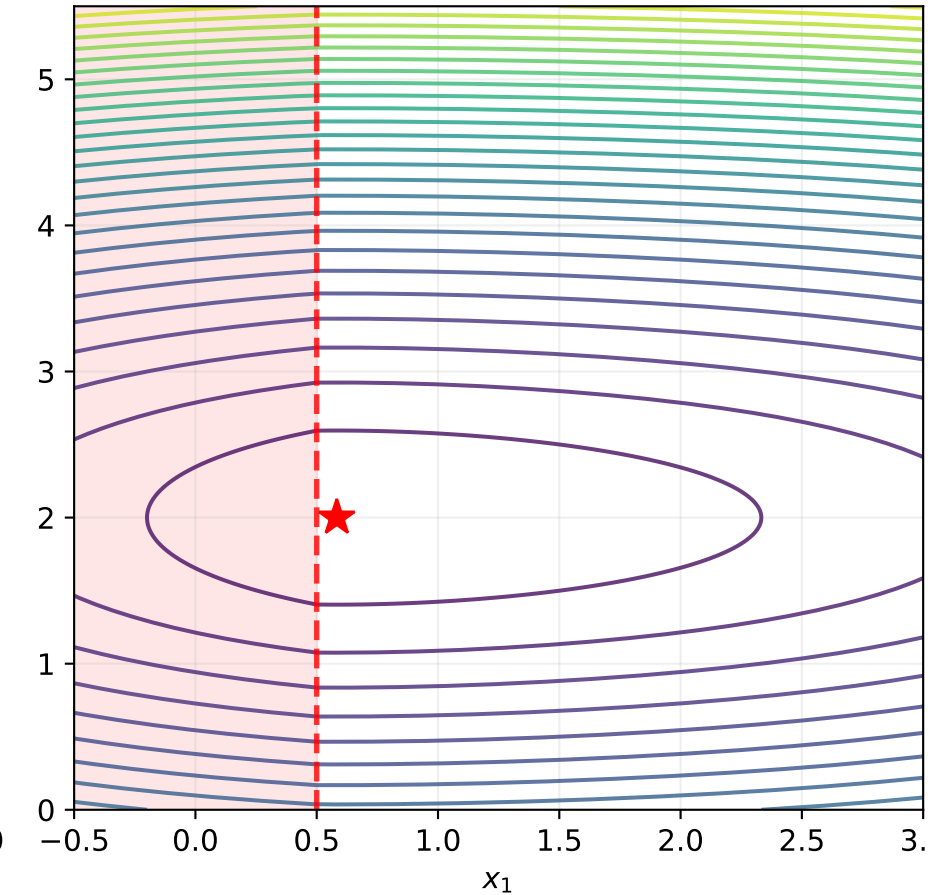
$\lambda = 0.0$
(unconstrained)



$\lambda = 0.15$



$\lambda = 1.8$



$\lambda = 4.5$

